



Probabilistic assessment of water network influence zones and ecosystem responses following the middle route of the South-to-North Water Diversion Project

Tianliang Jiang, Yanping Qu, Xuejun Zhang, Dandan Su, Yu Han & Ruixuan Zhao

To cite this article: Tianliang Jiang, Yanping Qu, Xuejun Zhang, Dandan Su, Yu Han & Ruixuan Zhao (2026) Probabilistic assessment of water network influence zones and ecosystem responses following the middle route of the South-to-North Water Diversion Project, Big Earth Data, 10:1, 400-423, DOI: [10.1080/20964471.2025.2597579](https://doi.org/10.1080/20964471.2025.2597579)

To link to this article: <https://doi.org/10.1080/20964471.2025.2597579>



© 2025 The Author(s). Published by Taylor & Francis Group and Science Press on behalf of the International Society for Digital Earth, supported by the International Research Center of Big Data for Sustainable Development Goals.



Published online: 10 Dec 2025.



Submit your article to this journal [↗](#)



Article views: 302



View related articles [↗](#)



View Crossmark data [↗](#)

RESEARCH ARTICLE



Probabilistic assessment of water network influence zones and ecosystem responses following the middle route of the South-to-North Water Diversion Project

Tianliang Jiang^{a,b}, Yanping Qu^{a,b}, Xuejun Zhang^{a,b}, Dandan Su^{c,d}, Yu Han^{a,b} and Ruixuan Zhao^{a,b}

^aState Key Laboratory of Water Cycle and Water Security, China Institute of Water Resources and Hydropower Research, Beijing, China; ^bResearch Center on Flood and Drought Disaster Reduction, China Institute of Water Resources and Hydropower Research, Beijing, China; ^cState Key Laboratory of Efficient Utilization of Arid and Semi-Arid Arable Land in Northern China, Chinese Academy of Agricultural Sciences, Beijing, China; ^dInstitute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences, Beijing, China

ABSTRACT

Large-scale inter-basin water transfer (IBWT) projects fundamentally reshape regional hydrological networks, triggering profound eco-hydrological consequences. This study quantitatively assesses the structural transformation of water networks and their ecohydrological consequences in central and southern Hebei Province, a major recipient area of the Middle Route of China's South-to-North Water Diversion (SNWD) project. By integrating multi-source remote sensing data, environmental variables, and machine learning techniques, we developed a probabilistic framework to delineate hydrological influence zones and their associated seasonal dynamics before and after the project's implementation. Utilizing water network density and connectivity indices derived from high-resolution surface water datasets, we quantified changes in network structure across 175 water distribution units. Our findings indicate a 23% increase in water network density and a 26.7% improvement in connectivity, reflecting a substantial enhancement of hydrological integration post-2015. Land use classification models (overall accuracy: 91.7%) identified water network density and soil moisture as primary determinants of ecosystem distribution, with forests and grasslands exhibiting heightened sensitivity to seasonal hydrological variations. The newly constructed artificial water conveyance pathways significantly expanded the spatial extent of ecological influence zones, particularly during the wet season. Probabilistic modeling further revealed distinct environmental preferences among land cover types, and calibrated ecological influence zones closely aligned with existing irrigation district boundaries. These findings underscore the importance of

ARTICLE HISTORY

Received 17 July 2025

Accepted 6 November 2025

KEYWORDS

Inter-basin water transfer; water network structure; eco-hydrological impact; South-to-North Water Diversion Project; Hebei Province

CONTACT Yanping Qu  quyp@iwhr.com  State Key Laboratory of Water Cycle and Water Security, China Institute of Water Resources and Hydropower Research, Beijing 100038, China

© 2025 The Author(s). Published by Taylor & Francis Group and Science Press on behalf of the International Society for Digital Earth, supported by the International Research Center of Big Data for Sustainable Development Goals.

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

dynamic, data-driven approaches for assessing the ecological consequences of engineered hydrological changes. The proposed framework offers a scalable methodology for evaluating the ecological consequences of IBWT projects, thereby supporting the development of more responsive and ecologically attuned water resource management strategies.

1. Introduction

Water networks—comprising rivers, streams, lakes, and wetlands—constitute the foundational circulatory systems of watershed ecosystems (Leibowitz et al., 2018; Wang et al., 2022). These hydrological systems are fundamental to the transport of water, nutrients, energy, and organisms, thereby sustaining the ecological integrity and functionality of terrestrial and aquatic environments (Wang et al., 2022; Wei et al., 2023a). Riparian forests and wetland vegetation, for instance, typically align along flow paths and exhibit adaptive differentiation in response to moisture gradients (Friedman et al., 2022; Park & Kim, 2020). The distribution and productivity of ecosystems such as forests, grasslands, and shrublands are fundamentally governed by water availability, and their physiological traits reflect long-term adaptation to local hydrological conditions (Su et al., 2023). Similarly, agricultural ecosystems rely critically upon irrigation services provided by river networks, with crop yields contingent upon the spatial and temporal allocation of water resources (Du et al., 2024). Consequently, structural alterations to water networks can disrupt water accessibility and flow regimes, leading to significant changes in vegetation composition, land cover types, and overall ecological functioning (Zhang et al., 2023).

Inter-basin water transfer (IBWT) projects, widely employed to address spatial mismatches in water availability, can profoundly reconfigure the dynamics of river systems (Dong et al., 2023). These projects introduce artificial conveyance systems, such as canals, tunnels, and pipelines, creating new hydrological pathways and redistributing surface water across basins (Dong et al., 2023; Zhang et al., 2024). A prominent example is China's South-to-North Water Diversion (SNWD) Project, which transfers water from the Yangtze River to the North China Plain—a region critical for national food security that confronts severe water scarcity, exacerbated by historical groundwater over-extraction and ecological degradation (Kattel et al., 2020; Zhao et al., 2023). Such large-scale interventions not only re-engineer surface water hydroperiods by changing flow magnitude, timing, and duration (Yan et al., 2023) but also alter the complex interplay between surface water, groundwater recharge, and soil moisture (Kattel et al., 2019, 2020). By reshaping local river networks and increasing waterbody density, the SNWD project has induced significant ecological consequences that demand rigorous evaluation (Yan et al., 2023).

Quantifying the ecological implications of altered river networks is challenging due to the intricate interplay between hydrological changes and ecological responses (Newcomb & Godsey, 2023). In the context of large-scale IBWTs, two central questions emerge: How can we precisely delineate the spatial footprint of a water network's influence, and how can we effectively characterize its seasonal variability? Answering these questions is crucial for advancing eco-hydrological assessments and informing adaptive management strategies for engineered water transfer systems.

Seasonal dynamics of river networks are key drivers of ecosystem processes, typically manifested through fluctuations in streamflow, variations in surface water extent, and changes in groundwater tables (Johnson et al., 2023). Prior research on ecological influence zones has concentrated on several focal areas: (1) delineating riparian buffer zones (Lind et al., 2019); (2) identifying floodplains through hydraulic modeling (Pizzileo et al., 2024); (3) examining hyporheic zones (Morén et al., 2023); and (4) mapping moisture gradients based on species distribution (Boisseaux et al., 2025). However, most existing studies are limited to individual river reaches or small watersheds and lack broader regional evaluations, particularly in landscapes characterized by complex terrain and diverse land-use patterns.

This gap is especially problematic for IBWT projects, which reconfigure water networks at large spatial scales. While many studies have examined such projects from the perspective of interannual water balance, their seasonal eco-hydrological consequences, including impacts on the coupled surface water-groundwater-soil moisture system, remain insufficiently explored (Kattel et al., 2020; Li et al., 2023; Missik et al., 2021). Overlooking these dynamics limits our ability to evaluate the full extent of hydrological influence and hinders the development of seasonally responsive operational plans and ecological compensation strategies.

Moreover, conventional methods for delineating ecological influence zones—such as fixed-width buffer analyses—often fail to capture the dynamic spatial variability driven by seasonal hydrological conditions and heterogeneous landscapes (Nsenga Kumwimba et al., 2023; Swartz et al., 2024). Such static models can produce incomplete or biased assessments. In contrast, emerging data-driven methodologies offer promising alternatives. Machine learning (ML) techniques, for example, provide flexible means for modeling complex, nonlinear relationships between environmental drivers and ecosystem responses by processing high-dimensional data to predict land cover types, habitat suitability, and ecological changes (Vulova et al., 2025; Yu et al., 2025).

Leveraging these advanced data-driven approaches, this study addresses four primary objectives: (1) to evaluate the structure evolution of water networks in south-central Hebei Province and quantify changes before and after the implementation of the middle route of SNWD project; (2) to assess the influence of environmental drivers—especially hydrological factors—on ecosystem distribution; (3) to develop a probabilistic modeling framework for delineating ecological influence zones and their associated seasonal dynamics; and (4) to examine the differential responses of major ecosystem types (cropland, forest, grassland) to hydrological changes, with an emphasis on seasonal sensitivity. By addressing these objectives, this study contributes to a more integrated understanding of eco-hydrological interactions in the context of engineered water redistribution. The findings aim to support region-specific eco-hydrological assessments and inform the development of adaptive, ecologically grounded water resource management strategies in water-scarce regions.

2. Study area

The middle route of the SNWD project originates at the Taocha Canal Head of the Danjiangkou Reservoir in southern China, traversing Henan and Hebei provinces before reaching Beijing and Tianjin (Wu et al., 2021). Spanning a total length of 1,432 kilometers, the route intersects more than 200 rivers—including the Tangbai,

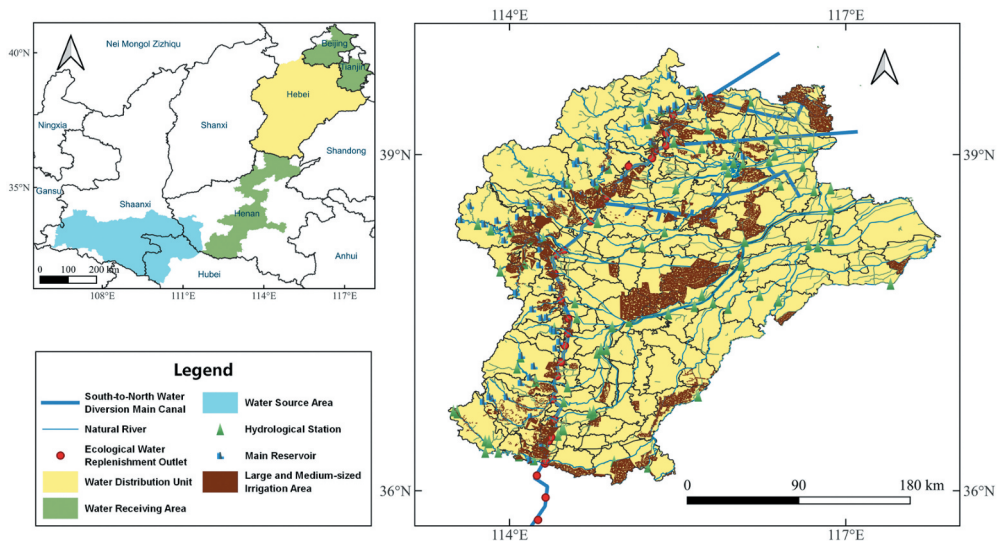


Figure 1. Spatial distribution of the water network and ecological infrastructure in the central and southern Hebei region under the SNWD project.

Shaying, Weihe, Fuyang, Futuo, and Juma Rivers—and supplies water to over 100 counties and 19 large- and medium-sized cities across four administrative regions (Figure 1). Since the completion of Phase I in December 2014, the Middle Route has delivered approximately 10.637 billion cubic meters of supplemental water ecological replenishment to more than 50 rivers in northern China as of year-end 2024.

This study focuses on south-central Hebei Province, a key water-receiving region within the Middle Route corridor. Historically, this area has been highly susceptible to drought, with major events recorded in 1920, 1924, 1942, 1965, 1972, and 1997 (Yang & Yao, 2024). These episodes have caused substantial ecological degradation and placed long-term stress on regional water resources. The introduction of inter-basin water transfer infrastructure presents an opportunity to alleviate ecological drought and restore environmental conditions in this region.

South-central Hebei is an ideal case study for this research due to its extensive hydrological infrastructure and the availability of long-term, multi-source datasets. Ongoing monitoring initiatives have produced rich time series and high-resolution spatial data, providing a strong foundation for assessing eco-hydrological dynamics with high precision. The region's climatic gradient, varied land use types, and complex terrain further enhance its value as a representative case for evaluating the eco-hydrological impacts of large-scale water transfer systems (Yuan et al., 2019). With its spatial heterogeneity ranging from irrigated plains to forested uplands, this region ensures that findings from the study can inform broader assessments of ecological responses under engineered hydrological regimes. By focusing on this critical nexus of natural and artificial water infrastructure, our analysis contributes to a deeper understanding of how inter-basin transfers interact with local ecological patterns and environmental vulnerabilities.

3. Data and methods

3.1. Data

This study focuses on south-central Hebei Province, covering an area of approximately 88,600 km². It utilizes a suite of high-resolution spatial and temporal datasets to support the quantitative evaluation of water network structure, environmental drivers, and ecosystem responses in south-central Hebei Province. Surface water dynamics were assessed using the Joint Research Centre (JRC) Monthly Water History v1.4 dataset, which provides global surface water extent at a 30-meter resolution from 1984 to 2022 (Pekel et al., 2016). This dataset enabled long-term tracking of hydrological changes before and after the implementation of the SNWD project. Land cover data were derived from the 30-meter annual land use dataset developed by Yang and Huang (2021), which documents land cover changes across China from 1990 to 2022. To characterize subsurface hydrological processes and vegetation rooting behavior, the study incorporated maximum rooting depth estimates from Fan et al. (2017). Soil physical properties, including particle size distribution and texture, were obtained from the Regridded Harmonized World Soil Database v1.2 (Wieder, 2014). This product comprises 27 NetCDF files at a 0.05-degree spatial resolution and is widely used in land surface and ecosystem process modeling. Climatic variables, including monthly precipitation and temperature, were extracted from the Climatic Research Unit (CRU) TS dataset (Harris et al., 2020). This gridded product is suitable for long-term eco-hydrological assessments and has been extensively validated across China's climatic zones.

3.2. Data preprocessing and harmonization

To ensure spatial consistency across the multi-source datasets, a rigorous harmonization process was implemented. All spatial datasets were reprojected to a common coordinate system (WGS 84/UTM Zone 50N) and resampled to a uniform 30-meter spatial resolution to match the finest-resolution layers (JRC surface water and land cover). Bilinear interpolation was employed for continuous variables (e.g., elevation, temperature, soil properties) to maintain smooth gradients, while the nearest neighbor method was applied to categorical data (e.g., land cover) to preserve class integrity. This standardized data stack formed the analytical foundation for all subsequent modeling.

3.3. Analytical framework

This study establishes a structured technical framework comprising three main phases (Figure 2). The analytical framework focuses exclusively on the eco-hydrological impacts of the SNWD Middle Route. The analytical framework focuses exclusively on the eco-hydrological impacts of the SNWD Middle Route. While the planned Phase II of the Eastern Route is expected to influence this region, it was not yet operational during the study period. Existing water transfers associated with the Eastern Route's alignment, such as the emergency replenishment to Baiyangdian Lake, are intermittent and involve small volumes diverted from the

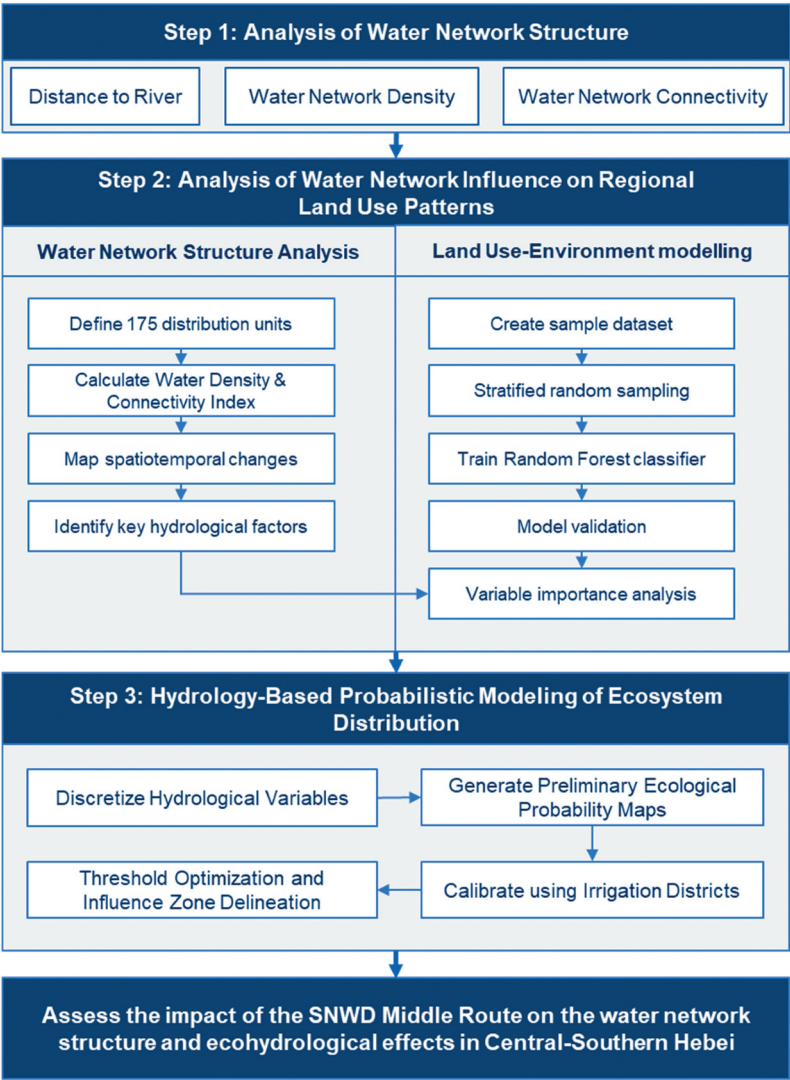


Figure 2. Technical framework diagram of the study.

Yellow River. Their impact is considered negligible compared to the large-scale, continuous flows from the Middle Route, which are the subject of this investigation.

The first phase involves foundational data preparation through the acquisition and harmonization of multi-source datasets as previously detailed. In the second phase, a parallel analytical strategy is employed to investigate the eco-hydrology impacts of the SNWD Middle Route Project on regional eco-hydrology, focusing on changes in water network structure and the relationships between land use and environmental variables. The final phase integrates key hydrological factors to develop a probabilistic model for delineating ecological influence zones based on joint conditional probabilities.

3.3.1. Analysis of water network structure

To evaluate structural changes in the regional water network before and after the implementation of the SNWD project (Phase I, completed in 2014), this study utilized remote sensing-derived binary raster datasets representing pre- and post-diversion water areas, extracted from the JRC monthly water history dataset. To capture seasonal dynamics, monthly surface water layers were aggregated into seasonal composites. While the JRC dataset algorithm inherently minimizes artifacts from cloud cover and topographic shadows through its long-term temporal aggregation; all seasonal composites still underwent a visual quality assessment to ensure data integrity before analysis.

Water network density (WD) was calculated across 175 water distribution units in central-southern Hebei Province. These units were derived by intersecting Level-3 administrative boundaries with Level-3 water resource zoning, ensuring that both governance and hydrological contexts were represented. For each unit, the number of water-classified pixels was aggregated and converted into surface water area. WD was then computed as:

$$WD = A_w/A_u \quad (1)$$

Where A_w represents total area classified as water within the unit, and A_u represents total area of the distribution unit. This metric provides a standardized measure of water presence across space and time.

To assess internal water network integrity, a connectivity index (CI) was calculated. This index characterizes the degree of fragmentation in the water network (Wei et al., 2023b). Within each unit, spatially contiguous water pixels were grouped into connected components using an 8-neighbor rule. The total number of such components (N_c) was recorded and compared to the total number of water pixels (N_w). The connectivity index was defined as:

$$CI = N_c/N_w \quad (2)$$

A lower CI value indicates a more continuous and integrated water network, while a higher value reflects fragmentation and lower hydrological cohesion.

3.3.2. Analysis of water network influence on regional land use patterns

To quantitatively assess how eco-hydrological variables influence land use patterns, a machine-learning-based classification framework was developed. This framework integrates remote sensing data, environmental predictors, and classified land use maps to model the relationships between land cover categories and eight environmental factors. These variables include Distance to River (DR), Clay Content (CC), Root Depth (RD), Connectivity Index (CI), Digital Elevation (DE), Soil Moisture (SM), Average Temperature (AT), and Water Network Density (WD).

3.3.2.1. Sample construction and target variable encoding. The target variable, land use type, was extracted from a classified raster product, which included six main categories: cropland, forest, grassland, shrubland, water bodies, and impervious surfaces. To minimize computational burden and reduce class imbalance, stratified random sampling was applied to select a total of 100,000 samples. This method ensured proportional representation across all land use categories by preserving their original distribution. While this approach maintains the natural

class distribution, it does not explicitly correct for imbalance. The Random Forest classifier was selected in part for its inherent robustness to moderate class imbalance, reducing the need for data augmentation techniques that could introduce artificial patterns. Each land use class was encoded as an integer to serve as the response variable for supervised classification. The dataset was partitioned into training and testing subsets using a 70:30 split, with the class distribution preserved in both subsets to maintain representativeness.

3.3.2.2. Model training and interpretation. A Random Forest (RF) classifier was chosen due to its ability to model nonlinear relationships, handle multicollinearity, and manage high-dimensional input data. The model was implemented using the scikit-learn library with 100 decision trees and default parameters. To interpret model behavior, variable importance scores were computed based on the mean decrease in impurity (Gini index). These scores quantify the relative contribution of each predictor in reducing classification error (Trok et al., 2023).

3.3.3. Hydrology-based probabilistic modeling of ecosystem distribution

To further examine how hydrological structures influence ecosystem distribution, a probabilistic modeling approach was developed using three key hydrological indicators: DR, WD, and CI. These variables were selected based on their high importance scores in the land use classification model and their direct relevance to surface water accessibility and network structure.

3.3.3.1. Generation of preliminary ecological probability maps. Each of the three hydrological variables (DR, WD, CI) was discretized into k equal-width intervals. For example, the domain of DR was partitioned into intervals. This process resulted in a three-dimensional environmental space composed of $k \times k \times k$ joint intervals, denoted as I_{ijk} . Each pixel in the study area was assigned to a specific joint interval based on its DR, WD, and CI values.

$$I_{DR} = \{I_{DR,1}, I_{DR,2}, \dots, I_{DR,k}\} \quad (3)$$

3.3.3.2. Generate preliminary ecological probability maps. For each land use category L , the number of pixels falling into each joint interval I_{ijk} was counted as $N_{L,ijk}$. The total number of pixels in that interval, regardless of class, was denoted as N_{ijk} . The conditional probability of observing land use type L under a given hydrological condition I_{ijk} was then computed using:

$$P(L|I_{ijk}) = \frac{N_{L,ijk}}{N_{ijk}} \quad (4)$$

This conditional probability reflects the environmental preference of each ecosystem type under combined hydrological scenarios. By assigning these probabilities back to their corresponding spatial pixels, preliminary ecological probability maps were generated for each land use category. These maps describe the likelihood of ecosystem occurrence based solely on hydrological structure and proximity.

3.3.3.3. Calibration using irrigation districts. To address potential underestimation or overgeneralization in the raw probability maps, a calibration step was introduced using known irrigation district boundaries as reference zones. Given that long-term irrigation in the North China Plain is a dominant factor in maintaining regional hydrological balance and sustaining both agricultural and non-agricultural ecosystems, these districts serve as an ecologically meaningful proxy for areas under maximum, consistent hydrological influence. The validity of these reference zones was further confirmed by cross-referencing their boundaries with official governmental water resource planning maps from the Hebei Provincial Department of Water Resources. These districts represent areas with long-term, engineered water delivery, making them a reliable proxy for lands under maximum hydrological influence. Pixels located within irrigation districts were assumed to be fully influenced by the water network and thus assigned a reference probability value of 1. A calibration function was fitted based on the relationship between the initial model-derived probabilities and the assumed ideal probability within these irrigation zones. This function adjusted the full probability surface across the study area, aligning modeled values more closely with observed water-use patterns.

3.3.3.4. Threshold optimization and influence zone delineation. Following calibration, a series of candidate probability thresholds (e.g., 0.70, 0.75, 0.80, 0.85, 0.90) were tested to delineate the final ecological influence zones. For each threshold, the spatial overlap between the extracted ecological influence zone and the reference irrigation districts was computed. The optimal threshold was selected by maximizing the intersection area while minimizing the inclusion of regions outside the known irrigation zones.



Figure 3. Confusion matrix of land use simulation validation results based on stratified sampling in the central-southern Hebei region.

This approach ensured that the extracted ecological influence zones were ecologically meaningful and spatially consistent with water-dependent land use. The final delineated zones provide a seasonally adaptable, probabilistically derived representation of hydrological influence on ecosystems.

4. Results and analysis

4.1. Water network influence on regional land use patterns

The validation of the land use classification model demonstrated high predictive accuracy across the central-southern Hebei region. As shown in the confusion matrix (Figure 3), the model achieved an overall accuracy of 91.7% and a Kappa coefficient of 0.86, indicating substantial agreement beyond chance. Cropland, the most prevalent land use type, was classified with 92.6% accuracy. Impervious surfaces were classified with the highest category-specific accuracy at 98.4%, effectively capturing urban expansion. Forest areas attained 83.2% accuracy, while grasslands and water bodies recorded accuracies of 83.4% and 85.6%, respectively. Shrubland classification was less accurate at 76.9%, likely due to its limited spatial extent and dispersed distribution.

The analysis of variable importance (Figure 4) reveals that environmental drivers exert differing levels of influence across land use categories. For cropland, the most influential variables were soil moisture (0.23), clay content (0.20), and average temperature (0.16), underscoring the significance of hydro-edaphic conditions for agricultural productivity.

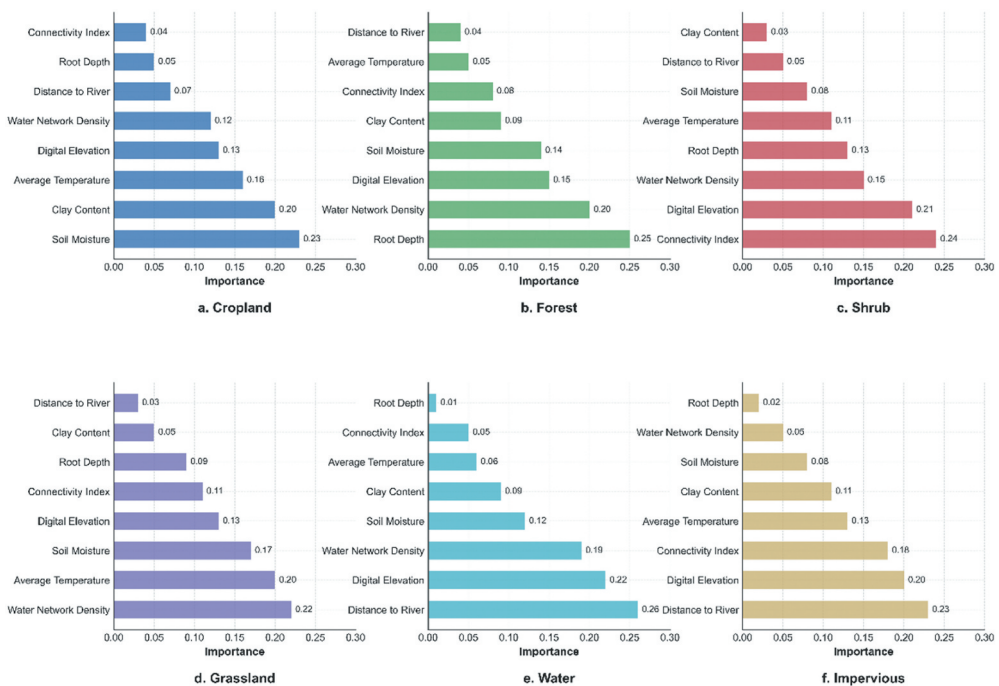


Figure 4. Relative importance of environmental variables for different land use types in the central-southern Hebei region.

Forest distribution (Figure 4(b)) was primarily shaped by root depth (0.25), water network density (0.20), and digital elevation (0.15), reflecting the importance of soil structure, moisture availability, and topography. Shrubland occurrence (Figure 4(c)) was most strongly associated with the connectivity index (0.24) and digital elevation (0.21). Grasslands (Figure 4(d)) showed high sensitivity to water network density (0.22), average temperature (0.20), and soil moisture (0.17), indicating a dependence on intermediate hydrological and climatic conditions. For water bodies (Figure 4(e)), the key variables were distance to river (0.26), digital elevation (0.22), and water network density (0.19). Finally, impervious surfaces (Figure 4(f)) correlated most strongly with distance to river (0.23), digital elevation (0.20), and the connectivity index (0.18).

From this analysis, several overarching patterns emerge. Variables quantifying water proximity and distribution, such as distance to river and water network density, consistently rank as dominant predictors. DR is paramount for defining both water bodies and impervious surfaces, while water network density is a top-tier driver for Grassland, Forest, and Water. In parallel, the importance of landscape structure is highlighted by the Connectivity Index, which is the single most critical factor for Shrubland and also highly significant for Impervious surfaces. Together, these key variables reveal that the spatial arrangement of water resources and the physical connectivity of the terrain are fundamental controls shaping land use patterns across the study area.

Probability distribution heatmaps (Figure 5) further illustrate these land use—environment interactions. Cropland exhibited peak probabilities of occurrence (0.60–0.70) at moderate water network densities (2–6 km/km²) and low elevations (<50 m). Forests were most likely to occur under conditions of deep rooting depth (>300 cm, probability = 0.81) and high connectivity indices (0.012–0.020, probabilities of 0.60–0.87). Elevation also played a key role, with optimal forest probabilities (0.45–0.58) found between 400–1000 m. Shrublands were concentrated in high elevation zones (>1,500 m, probability = 0.42). Grasslands displayed a relatively even distribution across environmental gradients. Water bodies were strongly associated with high water network density (16–20 km/km², probability = 0.80) and close proximity to rivers (0–100 m, probability = 0.65). Impervious surfaces were more likely to occur at moderate water network densities (6–8 km/km², probability = 0.05–0.08) and low to moderate elevations (50–100 m, probability = 0.10).

4.2. Water network changes in central-southern Hebei following the SNWD project

An analysis of water network structures across 175 distribution units highlights substantial hydrological transformation in central-southern Hebei Province following the 2015 implementation of the Middle Route of the SNWD project. As illustrated in Figure 6 spatial comparisons of pre- and post-diversion conditions (Figure 6(a,b)) reveal a clear expansion of surface water features. Network coverage increased significantly, with a 23% overall rise in mappable water extent, particularly in the eastern plains, including eastern Baoding, Langfang, and Cangzhou.

Water network density metrics (Figure 6(c,e)) show marked increases post-2015. The average water network density across all distribution units rose from 8.6% to 12.3%. Some eastern units, such as those in Cangzhou exhibited gains of over 18.7%. The spatial extent

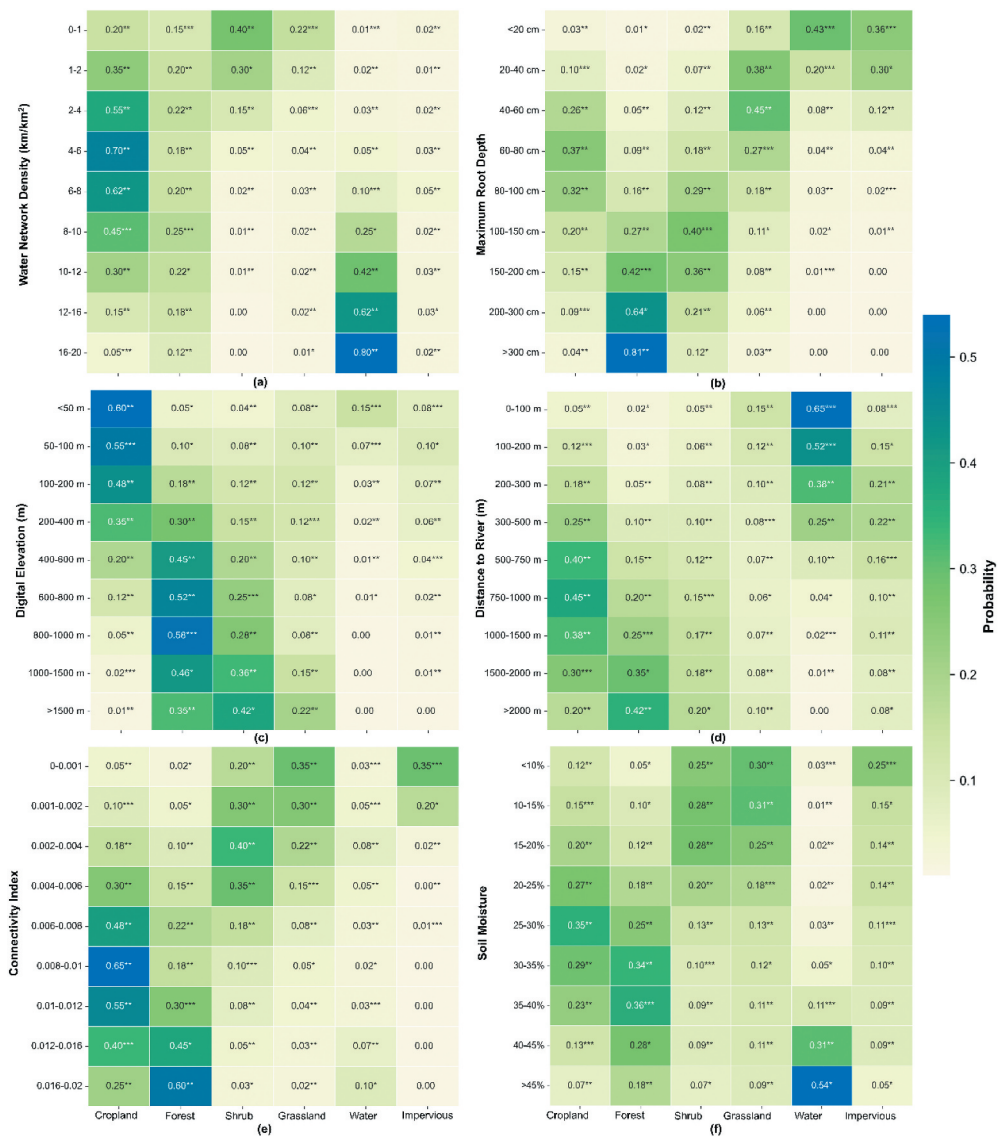


Figure 5. Probability distribution of land use types across various environmental factors in the central-southern Hebei region (significance levels: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$).

of high-density units (water coverage > 12%) nearly doubled, expanding from 18% to 27% of the region. This shift reflects the increased availability and redistribution of surface water enabled by the diversion infrastructure.

Changes in network cohesion are captured by the CI, as presented in Figure 6(d, f). The CI quantifies the degree of fragmentation within the water network; a lower value indicates that for a given water surface area, fewer, larger, and more continuous water bodies exist, thus signifying improved hydrological integration. After the diversion, the average CI declined from 0.015 to 0.011 across all units, indicating a 26.7% improvement in connectivity. Approximately 35% of all units

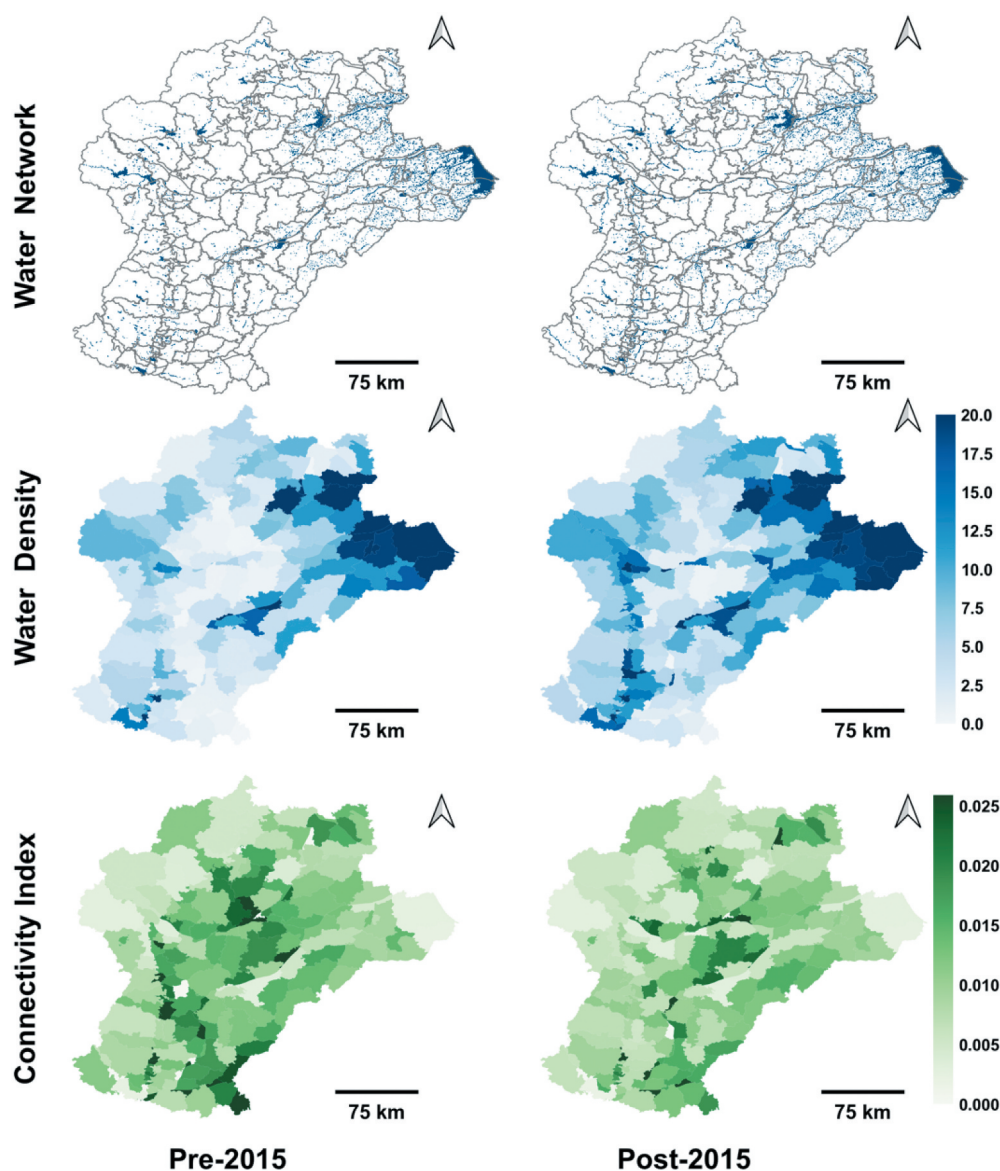


Figure 6. Spatial changes in water networks and connectivity in the central-southern Hebei region before and after the 2015 implementation of the SNWD Middle Route project.

experienced connectivity improvements exceeding 30%, with the most pronounced changes observed in central and eastern areas, such as the Ziya and Daqing River basins, which received direct inflows from the SNWD canal and its associated branches. These trends suggest that the SNWD Middle Route project enhanced both the density and the spatial continuity of surface water systems, transforming previously fragmented water bodies into a more cohesive and integrated network (Kattel et al., 2019, 2020).

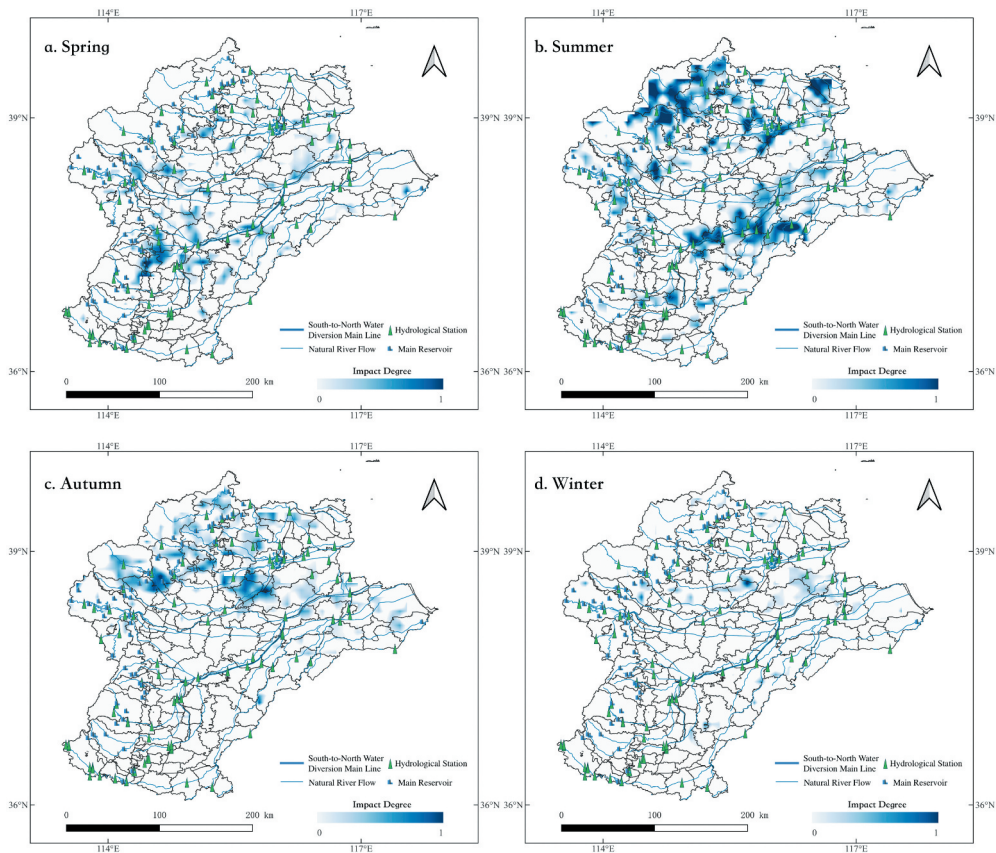


Figure 7. Degree of ecosystem impact in areas affected by water networks before 2015.

4.3. Spatiotemporal distribution of ecosystems affected by water networks

Before the implementation of the SNWD Middle Route, the eco-hydrological influence of the water network exhibited pronounced seasonal variation across central and southern Hebei. As shown in [Figure 7](#), the summer season was characterized by the most extensive and intense eco-hydrological effects, while winter recorded the most limited spatial extent. Analysis of land cover within these ecological influence zones ([Figure 8](#)) shows that cropland was the dominant land use. However, forest and grassland areas—which are particularly sensitive to hydrological fluctuations—expanded their footprint within the ecological influence zones during the wetter summer and autumn seasons, particularly in the northern and central upland subregions. In contrast, winter recorded the lowest levels of influence, with limited spatial extent and reduced intensity. Spring and autumn displayed moderate influence patterns, with localized impact clusters.

Following the commissioning of the SNWD Middle Route, the spatiotemporal dynamics of eco-hydrological influence underwent notable changes ([Figure 9](#)). The total area of the ecological influences zones expanded across all seasons compared to the pre-diversion period. The summer season ([Figure 9\(b\)](#)) remained the period of peak eco-hydrological influence, with the affected area increasing by 35.2%. These

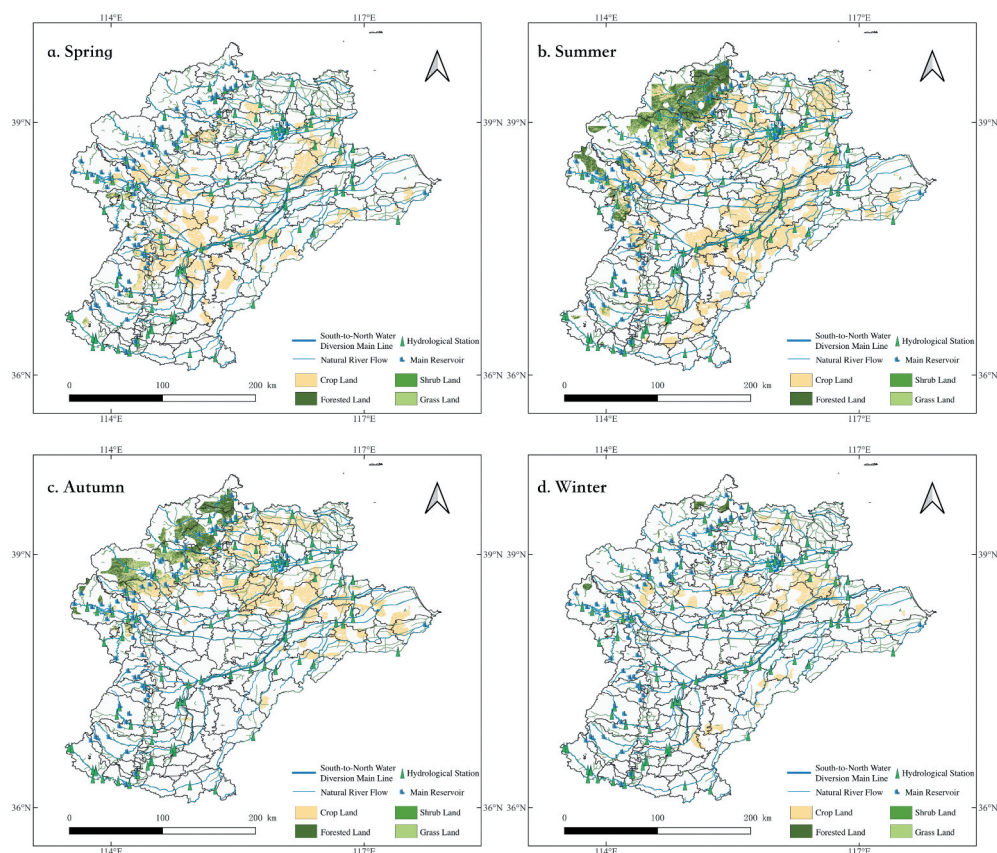


Figure 8. Ecosystem distribution within water network-affected areas before 2015.

zones were not only larger but also more spatially concentrated along the main canal and its adjacent corridors. The most substantial relative change occurred during the winter dry season (Figure 9(d)), where the ecological influence zone expanded by 45.4%. While the absolute area and impact intensity remained lower than in summer, this significant expansion demonstrates that the artificial conveyance system structurally altered seasonal flow regimes, sustaining ecological processes beyond the natural cycle. The shoulder seasons of spring (Figure 9(a)) and autumn (Figure 9(c)) also exhibited considerable expansion of their eco-hydrological footprints, with increases of 28.5% and 31.9%, respectively.

Figure 10 displays the corresponding land cover composition within the post-diversion ecological influence zones. Cropland continues to dominate, yet forests and grasslands show stronger spatial overlap with high-influence areas, particularly in summer and autumn. These ecosystems, primarily located in the northern uplands, have become increasingly responsive to the altered hydrological conditions. In summary, the Middle Route diversion has substantially altered the temporal and spatial characteristics of eco-hydrological interactions in the region, underscoring the need for adaptive ecological management that accounts for this new baseline of dependency on anthropogenic water delivery.

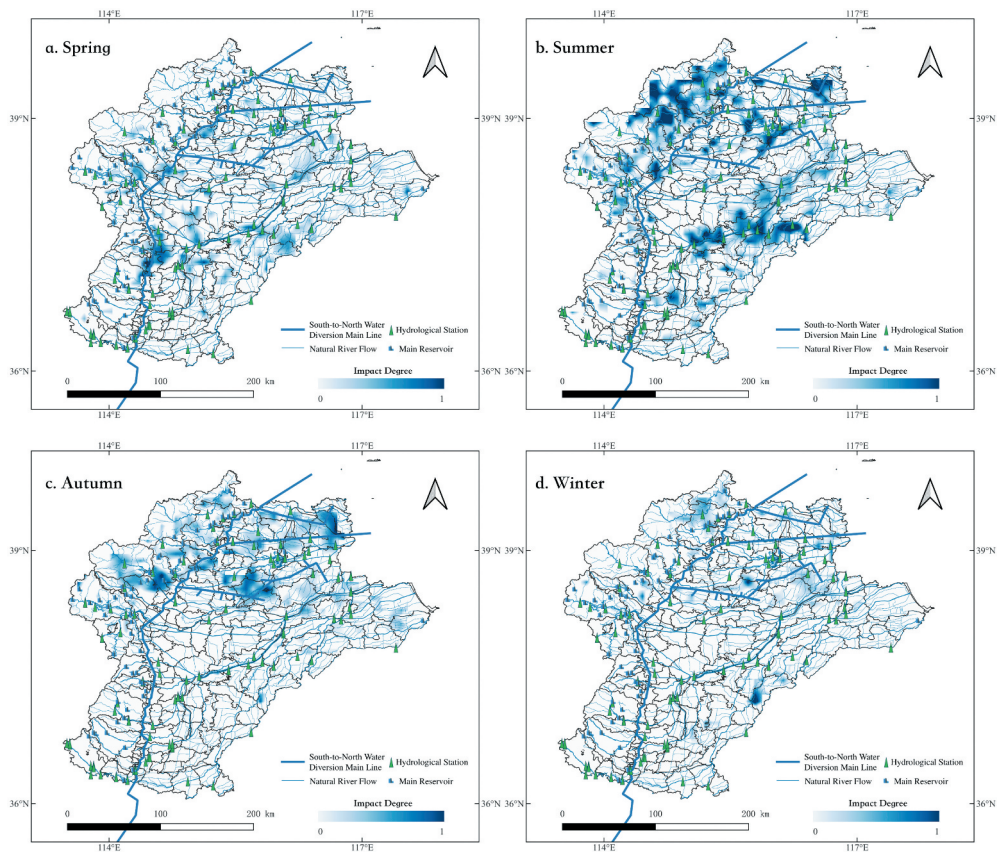


Figure 9. Degree of ecosystem impact in areas affected by water networks after 2015.

5. Discussion

5.1. Characteristics of post-diversion water network changes

Our analysis indicates that the SNWD Middle Route fundamentally reconfigured the hydrological landscape of central-southern Hebei. Post-diversion, surface water extent expanded by approximately 23%, and average water network density rose from 8.6% to 12.3%. The connectivity index improved by 26.7%, indicating that diverted flows linked previously isolated rivers, lakes, and reservoirs into a more continuous network, particularly in the eastern receiving areas.

The magnitude of this hydrological restructuring is substantial when contextualized with other large-scale IBWTs globally. For instance, while direct quantitative comparisons of connectivity metrics are rare, the observed expansion of surface water and enhanced integration in Hebei are comparable in scale to the eco-hydrological shifts reported following Spain's Tagus-Segura Water Transfer, which significantly altered floodplain hydrology and riparian ecosystems in the receiving Segura basin (Cañizares et al., 2022). The 26.7% improvement in network connectivity, in particular, signifies a profound enhancement of hydrological cohesion that exceeds the localized impacts often documented in smaller-scale diversions.

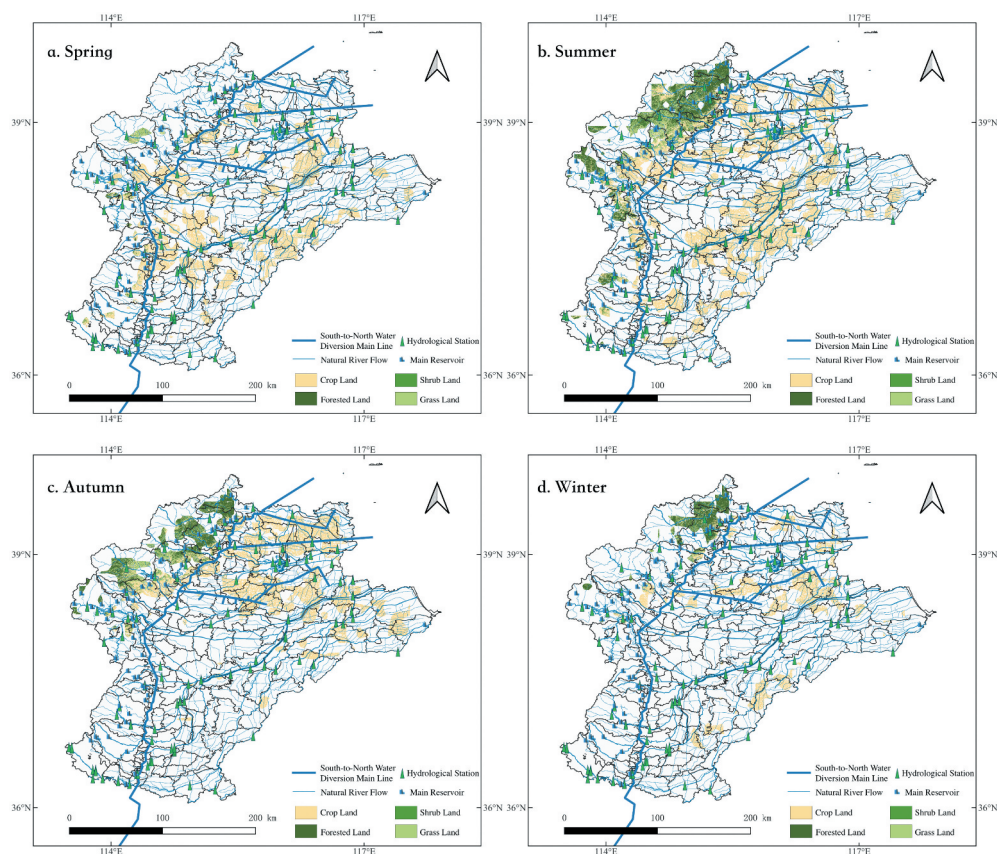


Figure 10. Ecosystem distribution within water network-affected areas after 2015.

5.2. Scope and key drivers of water network influence

The hydrological expansion driven by the diversion significantly widened the ecological influence zones of water networks. Post-diversion, affected areas along the main canal and its tributaries became larger and more contiguous, reflecting how augmented water flows propagated their influence across the landscape. Our analysis of predictor importance, which reveals that hydrological variables are dominant drivers of land cover patterns, aligns with the established principle that hydrological processes are key in shaping ecosystem structure and function (Gao et al., 2018).

Specifically, DR was a top predictor for water bodies and impervious surfaces, confirming that proximity to watercourses strongly governs moisture availability, riparian zonation, and development suitability (Stutter et al., 2021). WD was a highly influential factor for forests, grasslands, and water bodies, as denser networks enhance landscape-scale water availability and can create more favorable humid microclimates for vegetation (Stephens et al., 2021). The Connectivity Index was a key driver for multiple land covers, reflecting the vital role that hydrological connectivity plays in sustaining floodplain ecosystems and facilitating the transport of nutrients and organisms (Funk et al., 2023). Collectively, these results confirm that the spatial arrangement of water resources—

encapsulated by distance, density, and connectivity—is a principal control on land cover distribution in this region.

5.3. Seasonal dynamics and ecosystem-specific responses

Prior to the diversion, the water network's eco-hydrological influence exhibited strong seasonality, driven by the region's monsoonal climate, which concentrates water availability in the summer wet season (Deng et al., 2025). Consequently, summer had the broadest and most intense impact, whereas winter influence was minimal. After the Middle Route became operational, these seasonal patterns shifted. All seasons now exhibit larger and more contiguous ecological influence zones. Notably, the eco-hydrological influence during winter is substantially larger than before the diversion. Although its intensity remains lower than in summer, its persistence suggests that project flows now maintain aquatic and riparian conditions during the natural dry season, a key goal of supplemental water allocation strategies for ecological replenishment (Deng et al., 2025).

Throughout these changes, natural ecosystems (forests and grasslands) have remained far more sensitive to seasonal water variation than the region's dominant cropland. This differential sensitivity is consistent with findings that drought can severely degrade natural ecosystem productivity and increase forest mortality (Socha et al., 2023), whereas croplands remain relatively stable due to the buffering capacity of irrigation infrastructure (Zhu et al., 2021). This finding has critical implications for water resource management and ecological compensation. It suggests that strategies for ecological water replenishment should prioritize these sensitive natural ecosystems, as they stand to benefit most from targeted releases but are also most vulnerable to reductions in flow. Furthermore, the heightened responsiveness of forests and grasslands makes them valuable bio-indicators for monitoring the long-term ecological success of the water diversion project and for identifying areas under potential hydrological stress.

5.4. Limitations and future directions

This study introduces a probabilistic modeling framework, representing a significant methodological innovation for assessing the eco-hydrological impacts of IBWTs. Unlike static buffers that produce deterministic boundaries, our probabilistic approach quantifies the likelihood of eco-hydrological influence across a continuous spatial domain (Moe et al., 2021). This framework explicitly incorporates uncertainty and captures the dynamic interactions between multiple hydrological factors, thereby offering enhanced resolution, adaptability, and interpretability compared to traditional static models.

While the methodological framework proved effective, several limitations should be acknowledged, which in turn highlight promising directions for future research. The use of irrigation district boundaries as a calibration proxy for zones of maximum hydrological influence, for instance, has inherent constraints. These districts are engineered systems designed for agriculture and may not fully represent the nuanced hydrological responses of natural ecosystems. Although this approach was crucial for grounding our probabilistic model in real-world water management patterns, future studies could directly address this limitation by integrating high-resolution soil moisture data or vegetation water

content indices to better differentiate the hydrological responses of natural versus anthropogenic landscapes.

Building on the evaluation framework established in this study, future research could incorporate more advanced modeling techniques. The application of deep learning architectures and process-based ecosystem models (e.g., BIOME-BGC, LPJmL) offers promising avenues for improved prediction and interpretation, potentially providing a more comprehensive understanding of species-specific responses and biomass dynamics under changing hydrological regimes (Han et al., 2023; Li et al., 2023). Furthermore, extending the temporal scope of the analysis would yield deeper insights. Incorporating long-term remote sensing archives, such as the Landsat series, would allow for the reconstruction of historical ecosystem trends (Feng et al., 2016), while integrating finer resolution data from unmanned aerial vehicle (UAV) systems and field surveys of functional traits could validate model outputs and detail vegetation structure and eco-physiological responses (Spiers et al., 2025).

Collectively, these research directions will help overcome the limitations of the current study and contribute to a more systematic, data-rich assessment of the eco-hydrological impacts of IBWT projects, which is vital for developing adaptive management strategies that support both water security and ecological integrity.

6. Conclusion

This study presents a comprehensive analysis of the structural transformation of water networks and its eco-hydrological implications in central and southern Hebei Province following the implementation of the Middle Route of the SNWD project. By combining high-resolution remote sensing data, environmental variables, and advanced modeling techniques, we have developed a robust framework for evaluating the eco-hydrological consequences of large-scale IBWT initiatives.

First, our research demonstrates that the project induced substantial changes in the regional hydrological system. Quantitative indicators, including water network density and a connectivity index, revealed significant improvements in the spatial integration and continuity of surface water, particularly in the eastern and central subregions. These changes mitigated existing fragmentation and led to a more cohesive distribution of water resources.

Second, ecological responses to these hydrological changes were evident in both spatial patterns and seasonal dynamics. Our land use classification models identified water network density, soil moisture, and connectivity as the primary drivers of ecosystem distribution. Natural vegetation types, such as forests and grasslands, exhibited strong sensitivity to hydrological variation, while cropland distribution remained relatively stable due to irrigation.

Third, the proposed probabilistic modeling framework proved powerful in capturing the spatial footprint of eco-hydrological influence. By integrating multi-dimensional environmental gradients and calibrating model outputs with irrigation district boundaries, the resulting probability maps provided a sophisticated, spatially nuanced, and seasonally responsive alternative to static buffer-based methods.

Together, these findings underscore the profound eco-hydrological significance of engineered hydrological interventions. The framework developed is not merely an evaluative tool but also a practical instrument for adaptive management. Operationally, water managers can

use the probability maps to prioritize supplemental water allocations for ecological replenishment, optimizing the timing and magnitude of environmental flows to support sensitive natural ecosystems, particularly during dry periods. From a policy perspective, this research provides a quantitative basis for balancing agricultural and ecological water needs, enabling evidence-based decision-making that moves beyond simple water volume metrics to consider spatially explicit eco-hydrological outcomes. The framework's adaptability is rooted in its data-driven design; it can be transferred to other IBWT projects by incorporating region-specific remote sensing data and relevant environmental predictors, allowing for tailored assessments in different climatic and ecological contexts. In conclusion, this research provides critical insights into eco-hydrological processes within large-scale water infrastructure, offering a scalable methodology to support the development of more effective and ecologically attuned water resource management strategies in water-scarce regions worldwide.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This study was financially supported by the National Key Research and Development Program Funded Project (2023YFC3006604), Challenge Program of China Institute of Water Resources and Hydropower Research (JZ0145C142025), Water Conservancy Technology Demonstration Project (JZ110161B0032025), and the Special Project of Basic Scientific Research Business Expenses of China Institute of Water Resources and Hydropower Research (JZ110145B0032023).

Notes on contributors



Tianliang Jiang is a researcher at the China Institute of Water Resources and Hydropower Research (IWHR). He holds a Ph.D. degree, and his primary research interests focus on drought identification, drought propagation, and the ecological impacts of drought. As a sub-project leader and technical director, he has presided over two research topics within China's 14th Five-Year National Key R&D Program. He has also served as a key technical member in developing national and provincial drought monitoring and early warning platforms for regions including Anhui and Fujian. Dr. Jiang has authored over 20 papers in high-impact journals. In 2023, he was honored with inclusion in the Young Innovation Talent Promotion Plan by the Center for Disaster Reduction, IWHR.



Yanping Qu is a Professorate Senior Engineer and the Deputy Director of the Institute of Flood and Drought Disaster Reduction, China Institute of Water Resources and Hydropower Research (IWHR). She holds a Ph.D. degree and has received numerous accolades, including the "March 8th Red-Banner Pacesetter" award from the Central and State Organs and a personal commendation from the Ministry of Water Resources (MWR) in recognition of her outstanding contributions to water and drought disaster prevention. She is recognized as a "Scientific and Technological Talent" at IWHR and serves as a key member of the MWR's Innovative Talent Team for Strategic Research on Water and Drought Disaster Prevention. Dr. Qu has undertaken the role of Principal Investigator or Technical Director for more than 50 major research initiatives, including projects under

the National Key R&D Program. She served as the chief editor for the Implementation Plan for the First National Comprehensive Survey of Natural Disaster Risks (Drought Disaster) and spearheaded the development of the National Drought Monitoring and Early Warning Comprehensive Platform (Phase II). She has participated in formulating 6 national/industry standards, published over 90 academic papers, and authored 8 monographs. Her research achievements have been recognized with multiple awards, including two First Prizes of the Dayu Water Science and Technology Progress Award and a First Prize for Science and Technology from the Chinese National Committee on Large Dams.



Xuejun Zhang received his Ph.D. degree from the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences. He is currently a Professorate Senior Engineer at the China Institute of Water Resources and Hydropower Research. His research interests mainly focus on hydrometeorology, drought monitoring and forecasting, land surface modeling, and global change.



Dandan Su received her B.S. degree and M.S. degree in Agricultural Water and Soil Engineering from Gansu Agricultural University in 2024. She is currently working toward her joint Ph.D. degree in Soil Science at the Institute of Agricultural Resources and Regional Planning, Chinese Academy of Agricultural Sciences (CAAS), and the College of Resources and Environmental Sciences, Northwest A&F University, China. Her research interests focus on water relations in the soil-plant-atmosphere continuum and numerical modeling.



Yu Han received her B.Eng. degree in Agricultural Water Conservancy Engineering from Hohai University, China, in 2023. She is currently pursuing an M.Eng. degree in Water Disaster and Water Security at the China Institute of Water Resources and Hydropower Research (IWHR). Her research interests include drought risk assessment and management, environmental and water resources systems, and sustainable water management and ecosystem resilience.



Ruixuan Zhao received her B.E. degree in Hydrology and Water Resources Engineering from Yangtze University, China in 2021. She is currently working toward her M.S. degree in Water Disaster and Water Security at the China Institute of Water Resources and Hydropower Research, China. Her research interest is the identification and interpretation of water body information.

Data Availability Statement

The data that support the findings of this study are openly available in Mendeley Data at <https://doi.org/10.17632/njsj3dfsm3.1>.

AI disclosure statement

During the preparation of this work, the authors used Anthropic's Claude Sonnet 4 solely for language refinement, including improving clarity and adopting academic terminologies.

References

- Boisseaux, M., Nemetschek, D., Baraloto, C., Burban, B., Casado-Garcia, A., Cazal, J., Clément, J., Derroire, G., Fortunel, C., Goret, J.-Y., Heras, J., Jaouen, G., Maréchaux, I., Scoffoni, C., Vieilledent, G., Vleminckx, J., Coste, S., Schimann, H., & Stahl, C. (2025). Shifting trait coordination along a soil-moisture-nutrient gradient in tropical forests. *Functional Ecology*, 39(1), 21–37. <https://doi.org/10.1111/1365-2435.14679>
- Cañizares, A. O., Cantos, J. O., & Baños Castiñeira, C. J. (2022). The effects of climate change on the Tagus–Segura transfer: Diagnosis of the water balance in the Vega Baja del Segura (Alicante, Spain). *Water*, 14(13), 2023. <https://doi.org/10.3390/w14132023>
- Deng, H., Wang, Q., Zhao, Y., Zhu, Y., Jiang, S., Zhai, J., Gui, Y., Chen, X., Wang, L., & Liu, K. (2025). Impacts of continuous water diversions by the South-to-North Water Diversion project on increased precipitation and decreased temperature in water-receiving areas. *Geophysical Research Letters*, 52(7), e2024GL113549. <https://doi.org/10.1029/2024GL113549>
- Dong, J., Chen, X., Li, Y., Gao, M., Wei, L., Tangdamrongsu, N., & Crow, W. T. (2023). Inter-basin water transfer effectively compensates for regional unsustainable water use. *Water Resources Research*, 59(12), e2023WR035129. <https://doi.org/10.1029/2023WR035129>
- Du, D., Dong, B., Zhang, R., Cui, S., Chen, G., & Du, F. (2024). Spatiotemporal dynamics of irrigated cropland water use efficiency and driving factors in northwest China's Hexi Corridor. *Ecological Processes*, 13(1), 72. <https://doi.org/10.1186/s13717-024-00553-1>
- Fan, Y., Miguez-Macho, G., Jobbágy, E. G., Jackson, R. B., & Otero-Casal, C. (2017). Hydrologic regulation of plant rooting depth. *Proceedings of the National Academy of Sciences*, 114(40), 10572–10577. <https://doi.org/10.1073/pnas.1712381114>
- Feng, L., Han, X., Hu, C., & Chen, X. (2016). Four decades of wetland changes of the largest freshwater lake in China: Possible linkage to the Three Gorges Dam? *Remote Sensing of Environment*, 176, 43–55. <https://doi.org/10.1016/j.rse.2016.01.011>
- Friedman, J. M., Eurich, A. M., Auble, G. T., Scott, M. L., Shafroth, P. B., & Gibson, P. P. (2022). Response of riparian vegetation to short- and long-term hydrologic variation. *Ecological Applications*, 32(8), e2689. <https://doi.org/10.1002/eap.2689>
- Funk, A., Baldan, D., Bondar-Kunze, E., Brizuela, S. R., Kowal, J., & Hein, T. (2023). Connectivity as a driver of river-floodplain functioning: A dynamic, graph theoretic approach. *Ecological Indicators*, 154, 110877. <https://doi.org/10.1016/j.ecolind.2023.110877>
- Gao, H., Sabo, J. L., Chen, X., Liu, Z., Yang, Z., Ren, Z., & Liu, M. (2018). Landscape heterogeneity and hydrological processes: A review of landscape-based hydrological models. *Landscape Ecology*, 33(9), 1461–1480. <https://doi.org/10.1007/s10980-018-0690-4>
- Han, B. A., Varshney, K. R., LaDeau, S., Subramaniam, A., Weathers, K. C., & Zwart, J. (2023). A synergistic future for AI and ecology. *Proceedings of the National Academy of Sciences*, 120(38), e2220283120. <https://doi.org/10.1073/pnas.2220283120>
- Harris, I., Osborn, T. J., Jones, P., & Lister, D. (2020). Version 4 of the CRU TS monthly high-resolution gridded multivariate climate dataset. *Scientific Data*, 7(1), 109. <https://doi.org/10.1038/s41597-020-0453-3>
- Johnson, K., Harpold, A., Carroll, R. W. H., Barnard, H., Raleigh, M. S., Segura, C., Li, L., Williams, K. H., Dong, W., & Sullivan, P. L. (2023). Leveraging groundwater dynamics to improve predictions of summer low-flow discharges. *Water Resources Research*, 59(8), e2023WR035126. <https://doi.org/10.1029/2023WR035126>
- Kattel, G., Reeves, J., Western, A., Zhang, W., Jing, W., McGowan, S., Cuo, L., Scales, P., Dowling, K., He, Q., Wang, L., Capon, S., Pan, Z., Cui, J., Zhang, L., Xiao, L., Liu, C., Zhang, K., Gao, C. . . . Tian, Z. (2020). Healthy waterways and ecologically sustainable cities in Beijing–Tianjin–Hebei urban

- agglomeration (northern China): Challenges and future directions. *WIRs Water*, 8(2), e1500. <https://doi.org/10.1002/wat2.1500>
- Kattel, G. R., Shang, W., Wang, Z., & Langford, J. (2019). China's South-to-North Water Diversion project empowers sustainable water resources system in the north. *Sustainability*, 11(13), 3735. <https://doi.org/10.3390/su11133735>
- Leibowitz, S. G., Wigington, P. J., Jr., Schofield, K. A., Alexander, L. C., Vanderhoof, M. K., & Golden, H. E. (2018). Connectivity of streams and wetlands to downstream waters: An integrated systems framework. *JAWRA Journal of the American Water Resources Association*, 54(2), 298–322. <https://doi.org/10.1111/1752-1688.12631>
- Li, B., Sun, T., Tian, F., & Ni, G. (2023). Enhancing process-based hydrological models with embedded neural networks: A hybrid approach. *Journal of Hydrology*, 625, 130107. <https://doi.org/10.1016/j.jhydrol.2023.130107>
- Lind, L., Hasselquist, E. M., & Laudon, H. (2019). Towards ecologically functional riparian zones: A meta-analysis to develop guidelines for protecting ecosystem functions and biodiversity in agricultural landscapes. *Journal of Environmental Management*, 249, 109391. <https://doi.org/10.1016/j.jenvman.2019.109391>
- Missik, J. E. C., Liu, H., Gao, Z., Huang, M., Chen, X., Arntzen, E., Mcfarland, D. P., & Verbeke, B. (2021). Groundwater regulates interannual variations in evapotranspiration in a riparian semiarid ecosystem. *Journal of Geophysical Research: Atmospheres*, 126(7), e2020JD033078. <https://doi.org/10.1029/2020JD033078>
- Moe, S. J., Carriger, J. F., & Glendell, M. (2021). Increased use of Bayesian network models has improved environmental risk assessments. *Integrated Environmental Assessment and Management*, 17(1), 53–61. <https://doi.org/10.1002/ieam.4369>
- Morén, I., Mojarrad, B., Riml, J., & Wörman, A. (2023). Geographic and hydromorphologic controls on interactions between hyporheic flow and discharging deep groundwater. *Hydrogeology Journal*, 31(3), 537–555. <https://doi.org/10.1007/s10040-023-02599-5>
- Newcomb, S. K., & Godsey, S. E. (2023). Nonlinear riparian interactions drive changes in headwater streamflow. *Water Resources Research*, 59(10), e2023WR034870. <https://doi.org/10.1029/2023WR034870>
- Nsenga Kumwimba, M., Huang, J., Dzakupasu, M., De Silva, K., Ohore, O. E., Ajibade, F. O., Li, X., Jingjun, S., Muyembe, D. K., & Kaixuan, H. (2023). An updated review of the efficacy of buffer zones in warm/temperate and cold climates: Insights into processes and drivers of nutrient retention. *Journal of Environmental Management*, 336, 117646. <https://doi.org/10.1016/j.jenvman.2023.117646>
- Park, H., & Kim, J. G. (2020). Temporal and spatial variations of vegetation in a riparian zone of South Korea. *Journal of Ecology and Environment*, 44(1), 9. <https://doi.org/10.1186/s41610-020-00152-z>
- Pekel, J.-F., Cottam, A., Gorelick, N., & Belward, A. S. (2016). High-resolution mapping of global surface water and its long-term changes. *Nature*, 540(7633), 418–422. <https://doi.org/10.1038/nature20584>
- Pizzileo, S., Moretti, G., & Orlandini, S. (2024). Flood plain inundation modeling with explicit description of land surface macrostructures. *Advances in Water Resources*, 188, 104713. <https://doi.org/10.1016/j.advwatres.2024.104713>
- Socha, J., Hawryło, P., Tyimińska-Czabańska, L., Reineking, B., Lindner, M., Netzel, P., Grabska-Szwagrzyk, E., Vallejos, R., & Reyer, C. P. O. (2023). Higher site productivity and stand age enhance forest susceptibility to drought-induced mortality. *Agricultural and Forest Meteorology*, 341, 109680. <https://doi.org/10.1016/j.agrformet.2023.109680>
- Spiers, A. I., Scholl, V. M., McGlinchy, J., Balch, J., & Cattau, M. E. (2025). A review of UAS-based estimation of forest traits and characteristics in landscape ecology. *Landscape Ecology*, 40(2), 29. <https://doi.org/10.1007/s10980-024-01991-0>
- Stephens, C. M., Lall, U., Johnson, F. M., & Marshall, L. A. (2021). Landscape changes and their hydrologic effects: Interactions and feedbacks across scales. *Earth-Science Reviews*, 212, 103466. <https://doi.org/10.1016/j.earscirev.2020.103466>
- Stutter, M., Baggaley, N., Uallacháin, Ó. D., & Wang, C. (2021). The utility of spatial data to delineate river riparian functions and management zones: A review. *Science of the Total Environment*, 757, 143982. <https://doi.org/10.1016/j.scitotenv.2020.143982>

- Su, J., Zhang, Y., & Xu, F. (2023). Divergent responses of grassland productivity and plant diversity to intra-annual precipitation variability across climate regions: A global synthesis. *Journal of Ecology*, 111(9), 1921–1934. <https://doi.org/10.1111/1365-2745.14149>
- Swartz, A. G., Coble, A. A., Thaler, E. A., & Warren, D. R. (2024). Quantifying the variability of “fixed-width” buffers on harvested lands in western Oregon and Washington. *Journal of Forestry*, 122(5–6), 417–430. <https://doi.org/10.1093/jofore/fvae018>
- Trok, J. T., Davenport, F. V., Barnes, E. A., & Diffenbaugh, N. S. (2023). Using machine learning with partial dependence analysis to investigate coupling between soil moisture and near-surface temperature. *Journal of Geophysical Research: Atmospheres*, 128(12), e2022JD038365. <https://doi.org/10.1029/2022JD038365>
- Vulova, S., Horn, K., Rocha, A. D., Brill, F., Somogyvári, M., Okujeni, A., Förster, M., & Kleinschmit, B. (2025). Unraveling the response of forests to drought with explainable artificial intelligence (XAI). *Ecological Indicators*, 172, 113308. <https://doi.org/10.1016/j.ecolind.2025.113308>
- Wang, H., Huang, L., Hu, J., Yang, H., & Guo, W. (2022). Effect of urbanization on the river network structure in Zhengzhou City, China. *International Journal of Environmental Research and Public Health*, 19(4), 2464. <https://doi.org/10.3390/ijerph19042464>
- Wei, C., Wang, X., Cai, J., Liao, Z., Li, C., & Liu, Q. (2023a). A new framework combining hydrological connectivity metrics and morphological spatial pattern analysis for the hydrological connectivity evaluation of wetlands. *Integrated Environmental Assessment and Management*, 19(4), 1064–1078. <https://doi.org/10.1002/ieam.4708>
- Wei, C., Wang, X., Cai, J., Liao, Z., Li, C., & Liu, Q. (2023b). A new framework combining hydrological connectivity metrics and morphological spatial pattern analysis for the hydrological connectivity evaluation of wetlands. *Integrated Environmental Assessment and Management*, 19(4), 1064–1078.
- Wieder, W. (2014). RegridDED harmonized world soil database v1.2. *Analytical Chemistry*, 86(1), 826–834. <https://doi.org/10.3334/ORNLDAAAC/1247>
- Wu, Y., Li, L., Liu, Z., Chen, X., & Huang, H. (2021). Real-time control of the middle route of South-to-North Water Diversion project. *Water*, 13(1), 97. <https://doi.org/10.3390/w13010097>
- Yan, H., Lin, Y., Chen, Q., Zhang, J., He, S., Feng, T., Wang, Z., Chen, C., & Ding, J. (2023). A review of the eco-environmental impacts of the South-to-North Water Diversion: Implications for interbasin water transfers. *Engineering*, 30, 161–169. <https://doi.org/10.1016/j.eng.2023.05.012>
- Yang, G., & Yao, C. (2024). Drought hazards and hydrological variations in the South Hebei Plain of China over the past 500 years. *Atmosphere*, 15(10), 1243. <https://doi.org/10.3390/atmos15101243>
- Yang, J., & Huang, X. (2021). The 30 m annual land cover dataset and its dynamics in China from 1990 to 2019. *Earth System Science Data*, 13(8), 3907–3925. <https://doi.org/10.5194/essd-13-3907-2021>
- Yu, Z., Bu, C., & Li, Y. (2025). Machine learning for ecological analysis. *Chemical Engineering Journal*, 507, 160780. <https://doi.org/10.1016/j.cej.2025.160780>
- Yuan, D., Chen, Q., Shi, Y., Li, J., & Wang, S. (2019). Ecological security pattern of land use in a typical transect of Hebei Province. *Chinese Journal of Eco-Agriculture*, 27(11), 1767–1778. <https://doi.org/10.13930/j.cnki.cjea.190247>
- Zhang, H., Luo, J., Wu, J., & Dong, H. (2024). Dynamic multiobjective two-stage fuzzy stochastic strategy for optimal water allocation in inter-basin water division under changing environment: A case study of Hanjiang-to-Weihe water division in Shaanxi Province, China. *Journal of Cleaner Production*, 452, 142175. <https://doi.org/10.1016/j.jclepro.2024.142175>
- Zhang, X., Liu, Y., Zhao, W., Li, J., Xie, S., Zhang, C., He, X., Yan, D., & Wang, M. (2023). Impact of hydrological changes on wetland landscape dynamics and implications for ecohydrological restoration in Honghe National Nature Reserve, Northeast China. *Water*, 15(19), 3350. <https://doi.org/10.3390/w15193350>
- Zhao, Y., Guo, L., Li, Y., Zhu, Y., & Cheng, Z. (2023). Revealing the driving factors for structure changes of regional water consumption: An empirical study of the middle route project of the South-to-North Water Diversion in China. *Environment, Development and Sustainability*, 27(5), 10415–10438. <https://doi.org/10.1007/s10668-023-04316-4>
- Zhu, X., Liu, Y., Xu, K., & Pan, Y. (2021). Effects of drought on vegetation productivity of farmland ecosystems in the drylands of northern China. *Remote Sensing*, 13(6), 1179. <https://doi.org/10.3390/rs13061179>